

Section 12.4: The Cross Product

Vectors and the Geometry of Space

MTH 310
Calculus III

The Cross Product — A New Kind of Product

The **cross product** $\vec{a} \times \vec{b}$ takes two vectors in $\mathbb{R}^3$ and returns a **vector** that is **perpendicular** to both $\vec{a}$ and $\vec{b}$.

Quick Review: Determinants

2 $\times$ 2 Determinant

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

Example 1

Compute the determinant $\begin{vmatrix} 2 & 3 \\ 4 & 1 \end{vmatrix}$.

3 $\times$ 3 Determinant (Cofactor Expansion)

Expand along the **first row**:

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}$$

Notice the **alternating signs**: $+$, $-$, $+$.

Example 2

Compute the determinant:

$$\begin{vmatrix} 2 & 3 & 1 \\ 4 & 1 & -3 \\ -2 & 5 & 2 \end{vmatrix}$$

Expand along the first row using the cofactor formula.

The Cross Product Formula

Cross Product

If $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$, then:

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

The result is a **vector** (note the $\vec{i}, \vec{j}, \vec{k}$). This is a "symbolic" determinant — the first row contains vectors, not numbers.

Example 3: Computing a Cross Product

Example 3

Let $\vec{a} = \langle -1, 3, 4 \rangle$ and $\vec{b} = \langle 5, 2, 1 \rangle$. Find $\vec{a} \times \vec{b}$.

Set up the 3x3 determinant and expand along the first row.

Your Turn: Cross Product Practice

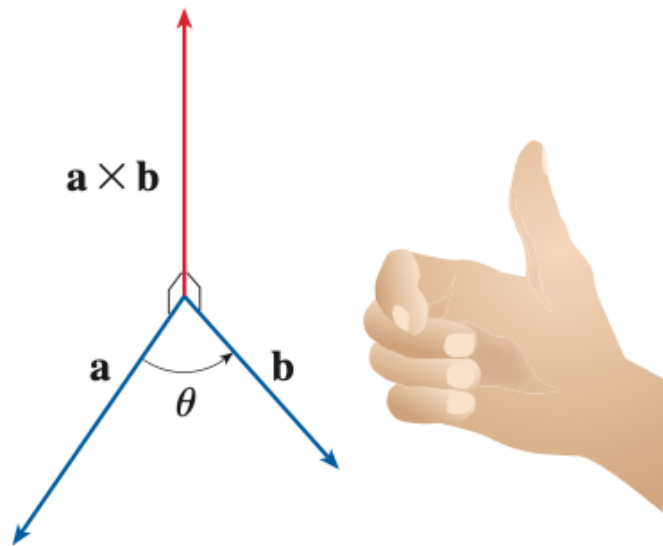
Example 4

Compute $\langle 1, 0, 2 \rangle \times \langle 3, 1, -1 \rangle$.

Set up the determinant and expand. Check your answer with a dot product.

Key Properties of the Cross Product

Order matters! $\vec{a} \times \vec{b} \neq \vec{b} \times \vec{a}$ (they point in **opposite** directions).



Magnitude of the Cross Product

Magnitude Formula

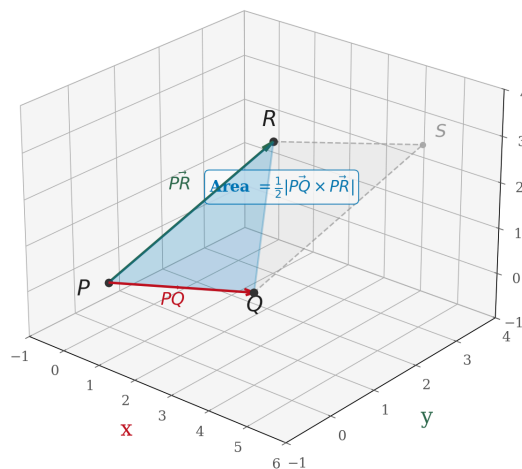
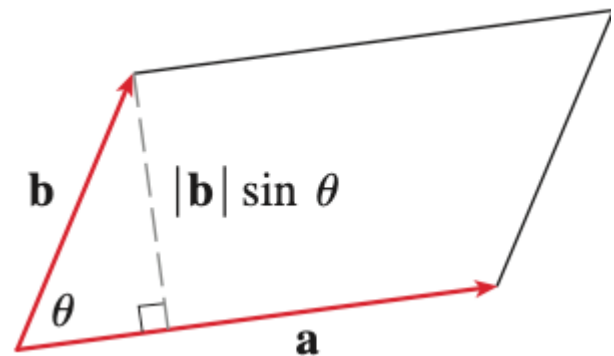
$$|\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin \theta$$

where θ is the angle between $\vec{a}$ and $\vec{b}$ ($0 \leq \theta \leq \pi$).

$\vec{a}$ and $\vec{b}$ are **parallel** if and only if $\vec{a} \times \vec{b} = \vec{0}$.

Area from the Cross Product

- **Parallelogram** with sides $\vec{a}$ and $\vec{b}$: Area = $|\vec{a} \times \vec{b}|$
- **Triangle** with sides $\vec{a}$ and $\vec{b}$: Area = $\frac{1}{2} |\vec{a} \times \vec{b}|$



Example 5: Normal Vector and Area from Three Points

Example 5

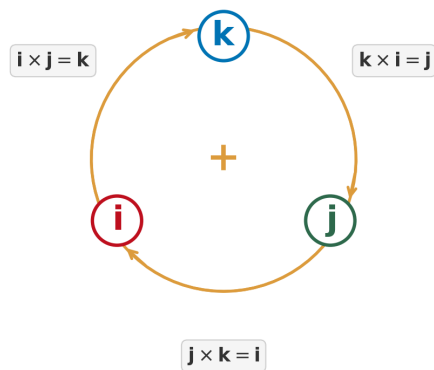
Given $P(-2, 1, 7)$, $Q(-3, 4, 11)$, and $R(3, 3, 8)$:

- (a) Find a vector perpendicular to the plane through P , Q , R .
- (b) Find the area of triangle PQR .

Form two edge vectors from P , then take their cross product.

Cross Products of the Standard Basis Vectors

Remember the cyclic pattern: $\vec{i} \rightarrow \vec{j} \rightarrow \vec{k} \rightarrow \vec{i} \rightarrow \dots$
Going forward gives a positive result; going backward gives a negative.



Homework

Section 12.4: 2, 3, 13, 16, 17, 19, 27, 29, 43, 44